NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_STAR CO_SC (MODEL-A & APEX) | Date: 04-01-2025
Time: 3 Hrs | GTM-11 | Max. Marks: 300

PHYSICS

1	**B**	2	**C**	3	**B**	4	**D**	5	**A**
6	**C**	7	**A**	8	**B**	9	**A**	10	**C**
11	**B**	12	**D**	13	**C**	14	**C**	15	**B**
16	**B**	17	**B**	18	**A**	19	**A**	20	**B**
21	**8**	22	**2**	23	**5**	24	**2**	25	**5**

CHEMISTRY

26	**D**	27	**B**	28	**B**	29	**D**	30	**B**
31	**B**	32	**D**	33	**D**	34	**D**	35	**D**
36	**D**	37	**C**	38	**D**	39	**C**	40	**B**
41	**A**	42	**A**	43	**D**	44	**D**	45	**B**
46	**6**	47	**50**	48	**2**	49	**195**	50	**500**

MATHS

51	**B**	52	**C**	53	**C**	54	**B**	55	**D**
56	**D**	57	**A**	58	**C**	59	**C**	60	**C**
61	**C**	62	**B**	63	**C**	64	**D**	65	**B**
66	**B**	67	**A**	68	**C**	69	**D**	70	**C**
71	**11**	72	**20**	73	**31**	74	**39**	75	**28**

For More Material Join: @JEEAdvanced_2025

PHYSICS

1. Ans: (B)

$\tau_{com} = 0$

2. Ans: (C)

Average momentum of gas molecules = 0 at all temperatures.

$$(V_{rms})_i = V, V_{rms} = \sqrt{\frac{3K_BT}{m}}, \quad T_f = 2T, m_f = \frac{m}{2} \rightarrow (V_{rms})_f = 2V$$

3. Ans :(B) $\bar{a}_{com} = \frac{m_1\bar{a_1} + m_2\bar{a_2}}{m_1 + m_2}, |\bar{a_1}| = |\bar{a_2}| = g$ vertically downwardly.

$\Rightarrow a_{com} = g$ independent of mass and speed.

4. Ans :(D)

a) Magnetic induction - $mT^{-2}A^{-1}$

b) Magnetic Flux - $mL^2T^{-2}A^{-1}$

c) Magnetic permeability - $mLT^{-2}A^{-2}$

d) Magnetization - $m^0L^{-1}A$

5. Ans: (A) Number of significant figures depends on precision and precision depends of least count of measuring instrument.

6. Ans: (C) Heat supplied to the system $\rightarrow$ dW +ve

Work done on the system $\rightarrow$ dW –ve

FLOT $dQ = dU + dW$

For this case $dQ = dU - dW$

7. Ans: (A)

Time taken by the wavelet from B to come out of the prism is more than time taken by wavelet from A

8. Ans: (B) $h = 0, when\, T = 0 \,\&\, \theta = 90^0$

9. Ans: (A) $i = \left(\frac{dQ}{dt}\right)_{conduction} = -KA\frac{dT}{dx}$

i $\rightarrow$ constant in stady scale

$\Rightarrow \frac{dT}{dx}$ = constant and –ve

10. Ans: (C) $E = mc^2$

When E decreases, m decreases.

11. Ans: (B) For paramagnetic substance. Magnetization increases when magnetic field increases.

When temperature increases magnetization decreases.

SAMAYEVA JAYATHE
THE NARAYANA GROUP
For More Material Join: @JEEAdvanced_2025

12. Ans: (D) Electromagnets less retention and less coercive force. (Should get easily demagnetized).

13. Ans: (C) Calorimeter does not take heat. So the internal energy of the system remains constant.

14. Ans: (C) The force on the left side of the centre is more.

15. Ans: (B) $V_{rms} = V_{avg}$

16. Ans: (B) Fringe width $\beta = \frac{\lambda d}{D}, \lambda_V < \lambda_R \Rightarrow \beta_V < \beta_R$

17. Ans: (B) Hole diffusion from p-side to n-side is due to broken bond on n-side and the freed electron from the bond jumps to broken bond on p-side.

18. Ans: (A) Drift current is due to motion of minority charge carriers due to electric field in p-n junction.

19. Ans: (A) Realising AND GATE by using diodes.

20. Ans: (B) Varies periodically with time.

21. $|L_o| = \frac{\sqrt{2}mu^3}{8g}, x = 8.$

22. $V_{string} = 2V_{com}$

$\therefore$ length through his hand $= 2l$, $x = 2$

23. $Y = \frac{Fl}{A\Delta l}$, slope of the graph $\tan 45^0 = \frac{\Delta l}{F} \Rightarrow \frac{\Delta l}{F} = 1$

$Y = \frac{l}{A} = 5 \times 10^4 N/m^2$, $x = 5$

24. $(f_0)_{closed} = \frac{V}{4l}$, $(f_0)_{open} = \frac{V}{2l}$

$(f_0)_{closed} = \frac{V}{4\left(\frac{l}{2}\right)} = \frac{V}{2l} = (f_0)_{open} \Rightarrow x = 2$

25. $\overline{L_0} = \left[\overline{L}_{translation}\right]_{com} + \left[\overline{L}_{rotation}\right]_{about\,com}$

$= mVR + \frac{2}{3}mR^2\omega = \frac{5}{3}mR^2\omega = \frac{5}{3}R^2\omega \Rightarrow a = 5.$

CHEMISTRY

26. Brown ring test is for identification of nitrate ion
27. Free radical reaction is favoured in gas phase and high temperature
28. The given test confirms the presence of Ni^{2+}
29. Reaction proceeds through radical mechanism
30. Bond angle in $Sncl_2$ is 96.5^0 while in SO_2 us 119.5^0
31. Q_{xy} and P_x orbitals favours the formation of $\pi - bond$

For More Material Join: @JEEAdvanced_2025

32. In isobaric expansion, magnitude of work done is same for both reversible and irreversible processes
33. All are correct statements
34. The probability of finding an electron at a point with in atom is proportional to $|\psi|^2$ at that point.
35. Kinetic energy of photo electron is related to energy of incident photon.
36. $Rate = Z_{AB} e^{-Ea/RT}$ predicts the rate constant for simple or elementary reactions only
37. Rate of both exothermic & endothermic reactions increase with temperature
38. Conceptual
39. Ninhydrin is used as a spraying reagents for proteins in paper chromatography.
40. Conceptual
41. NCERT purification chapter

50. For order reaction

$$k = \frac{\ln 2}{t_{1/2}} = \frac{0.7}{28 \times 60} = \frac{1}{40 \times 60} = \frac{1}{2400}$$

$$\log\left(\frac{1}{2400}\right) = 15 - \frac{9.19 \times 10^3}{T}$$

$$-2 - 1.38 = 15 - \frac{9.19 \times 10^3}{T}$$

$$\frac{9.19 \times 10^3}{T} = 18.38$$

$$T = \frac{10^3}{2} = 500K$$

TAMAYEVA JAYATHE
THE NARAYANA GROUP

For More Material Join: @JEEAdvanced_2025